

WELL COMPLETION REPORT (SYNTHETIC)

OIL-NHK-010 - Nahorkatiya-010

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000110
Field / District	Nahorkatiya / Dibrugarh
Block	PEL-ASSAM-02
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2009-03-24 / 2009-05-28
Measured / TVD	3263 m / 3222 m
Trajectory	Vertical; inc 3.4 deg; az 59.7 deg
Formation Interval	1409-2542 m
Lithology	Fine sandstone with carbonaceous shale
Pore Pressure Reference	3059 psi
Fracture Gradient	15.20 ppg
Mud System	Low-Solids WBM
NWIS Well Risk	Moderate (48.4/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	815	Surface
9 5/8 in casing	1892	Intermediate
7 in liner/casing	2871	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2009-04-09 948 m	Pressure Spike	High	Circulating pressure climbed without equivalent torque change. Mitigation: Use pressure trend thresholds referenced to depth and flow rate. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2009-04-12 1112 m	Pressure Spike	High	Standpipe pressure increased rapidly at nearly stable flow. Mitigation: Check solids loading and BHA hydraulics before drilling ahead. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.
2009-04-14 1194 m	Stuck Pipe	High	Torque and drag increased while ROP declined. Mitigation: Apply approved freeing procedure and avoid excessive overpull. Lesson: Avoid prolonged static periods in high-drag interval.
2009-05-01 2008 m	Torque Spike	High	High torque developed while WOB remained elevated. Mitigation: Reduce WOB/RPM and circulate before resuming normal parameters. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2009-04-20 1487 m	Torque Spike	Medium	Sharp torque excursion occurred with RPM instability. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Use offset torque trend to set adaptive operating envelope.
2009-04-22 1619 m	Torque Spike	Medium	High torque developed while WOB remained elevated. Mitigation: Monitor torque derivative rather than absolute torque alone. Lesson: Use offset torque trend to set adaptive operating envelope.
2009-04-29 1905 m	Mud Loss	Medium	Intermittent circulation losses developed while drilling ahead. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Track flow balance and pit volume trend before losses become severe.

Completion Summary

The synthetic well reached 3263 m MD in the Barail interval. The NWIS knowledge record links 11 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.